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| School of Electrical and Computer Engineering | <br>Shiraz University | Spring 1404             |
| Reinforcement Learning                        |                                                                                                        | Computer Assignment # 4 |

## Post Delivery Robot Simulation using TD Learning

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### Basic Setup and Environment

Basic setup and environment for the considered scenario in this computer assignment is exactly the same as the one in **computer assignment # 3**. However, we aim at solving the problem using several of the frameworks of temporal difference (TD) learning as explained in the following.

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### Part 1: Solve using Sarsa method

Implement a temporal difference learning control method to learn the optimal policy using the Sarsa method introduced in the course.

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### Part 2: Solve using Q-learning method

Implement a temporal difference learning control method to learn the optimal policy using the Q-learning method introduced in the course.

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### Part 3: Solve using $n$ -step TD methods

Implement temporal difference learning control methods to learn the optimal policy using  $n$ -step bootstrapping (set  $n = 4$ ). **In particular**, implement 4-step Sarsa and 4-step Q-learning methods introduced in the course.

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### Required Outputs (for 100,000 episodes):

1. Plot the **average return per episode overtime** for Parts 1-3.
2. Show the learned **optimal policy** by plotting the best action from each grid position. (You may need to report 4 figures for 4 possible package locations)
3. Plot the estimated **state-value function ( $V(s)$ )** after training is complete for each method. (You may need to report 4 figures for 4 possible package locations)

**Submission Requirements:**

1. The **source code** for the environment and agents.
2. Any plots you generate.
3. A **short report** explaining your approach and findings, and instructions on how to run your code.